OCTO-EARTH

Mission calculations · three individual enablers · separately listed support systems

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All numbers are model results of a physically hypothetical scenario, not engineering designs.

1. What the system is and the three enablers
Supporting architecture (not counted as enablers)
Chain of cause and effect
Key numbers at a glance
2. Physical constants and mission inputs
3. Mission requirement: how much push, energy and time
4. Enabler E1 — finite asteroid access; reaction-mass and launcher-support calculations
Selected enabler E1.1 - Finite asteroid-access assumption and reaction-mass reservoir
Supporting system E1.2 - Launch ring with 8 arms (mass driver)
Supporting system E1.3 - Arm power conditioning and energy reserve
Supporting system E1.4 - Directional arm radiators
Mars safety limits (control software, not an enabler)
Mercury, Venus and Jupiter (update p35; Saturn and beyond not modelled)
5. Enabler E2 — thin-film photovoltaic collector field; transfer-support calculations
Selected enabler E2.1 - Thin-film photovoltaic collector field
Supporting system E2.2 - Microwave phased-array transmitters
Supporting system E2.3 - Rectennas on the 8 arm tips
Supporting system E2.4 - Sail-tilt station keeping
Supporting system E2.5 - Clear sunlight lane
Supporting system E2.6 - Collector factories (Mercury material)
6. Enabler E3 — space-mirror ring and simplified climate
Selected enabler E3.1 - Space-mirror reflective field
Supporting system E3.2 - Terminator orbit and pointing
Climate model (what the mirrors must achieve)
7. Energy and mass ledger of the tested mission
8. Historical pre-hold comparison and the current control task
How the agent's five actions become physical commands
9. Selected enablers and supporting-system summary
What is NOT modelled (honest list)
10. What changes without a selected enabler or supporting system
11. Technical data sheets: selected enablers and supporting systems
12. Measuring instruments the control system needs (ordinary technology, not enablers)
13. Appendix: every parameter

1. What the system is and the three enablers

Historical results warning: the supplied comparison file predates the controlled-target-hold environment and has no matching training-environment version. Any results carried into this report are explicitly historical; they do not measure post-arrival maintenance. No current PPO retraining is being claimed.

OCTO-EARTH models Earth (with its Moon) moving from 1 AU toward a target circular orbit at **2.7 AU** over **10,000 simulated years**, through reaction-mass ejections: Earth recoils under momentum conservation. The three selected individual enablers are access to a finite asteroid (reaction-mass resource), a thin-film photovoltaic collector field (energy source), and a space-mirror ring (reflected light for the simplified climate model). Supporting launcher, power-transfer, radiator, factory and pointing equipment is listed separately; it is not bundled into those enablers. A PPO agent is one control option, but no current-version RL success is claimed here without an evaluation.

Evaluation is a two-part condition: the agent must actively hold the orbit and enabled climate within configured tolerances for 200 consecutive simulated years; an exit resets hold progress. This is a project setting, not an official rulebook threshold. A separate passive high-fidelity coast is then evaluated; it is not agent-controlled and does not train PPO.

Exactly three single enablers, not bundled systems. Supporting hardware is not thereby ordinary, available, safe or certified; its performance gaps are described independently. All reported energy sources, momentum exchange and orbital changes remain explicit in the model.

ID	Single selected enabler	Why selected	Scope, model representation and gap
E1	Access to a finite 3e22 kg asteroid	Without reaction mass the modeled recoil mechanism cannot transfer momentum to Earth.	A stated asteroid mass reservoir is available as reaction mass; the simulated launch packets are finite and Earth recoils when they are ejected. This is one resource-access assumption, not a launcher-system bundle. **Modelled:** Finite mass remaining, packet ejection, recoil, packet fates and mass/energy/momentum accounting. Hardware, capture and launch shock are not simulated; Earth-damage question P6 remains open. **Gap:** No asteroid is actually captured or docked; access to and placement of a 3e22 kg body is far beyond demonstrated capability.
E2	Thin-film photovoltaic collector field	The modeled mission needs a physical energy source; sunlight is traced through panel conversion and the downstream ledger.	Five million thin-film collectors at 0.5 AU intercept the configured 30 % of the Sun's output and convert sunlight into electricity. This is one power-generation enabler; microwave transmitters, rectennas and factories are supporting components, not bundled enablers. **Modelled:** Panel temperature and efficiency, degradation and replacement, build-up, manufacturing energy and mass, Goubau diffraction capture, Sun blocking, every loss in the energy ledger, clear lanes, the orbits of 12+12 followed collectors with / without station keeping (beam recoil, planets, light pressure). **Gap:** A Dyson-swarm-scale photovoltaic field (~1e13 x all solar panels ever built); collector area and mass are far beyond industry. The power-transfer chain separately assumes about 1e24 W over 0.5-3 AU, pointing near 1e-12 rad, and rectenna beam intensity around 2e8 times sunlight.

E3	Space-mirror ring	Without added reflected sunlight, the simplified climate model falls into a cold state at the target orbit.	A single mirror-ring technology reflects sunlight; the configured 1.6e15 m^2 film area can supply up to 1.10 x today's sunlight at 2.7 AU in the climate model. The agent sets the light level. **Modelled:** Capacity (1/r^2), light sent, light turned away, mirror heat (energy ledger), mass and the simplified climate effect. The mirrors' real orbits, light pressure, pointing errors and illumination pattern are not simulated. **Gap:** Plain reflection is known physics, but the configured 1.6e15 m^2 and formation flight/pointing are far beyond industry; this is not a physical deployment design.

Supporting architecture (not counted as enablers)

Associated with	Supporting equipment / assumptions
E1	• Launch ring + 8 launch arms (mass driver): Accelerates each packet to the commanded speed in the commanded direction; all 8 arms (each 2 R_E long, roots on a 1.25 R_E ring) throw one packet together. • Arm power conditioning + energy reserve: Turns the arm electricity into launch energy; a reserve covers launches when the beams are short. • Directional arm radiators: Reject the launcher and rectenna waste heat to space, away from Earth.
E2	• Microwave phased-array transmitter (one per collector): Turns the electricity into a focused 5.8 GHz beam aimed at the arms; suppresses side lobes so spill misses Earth. • Rectenna on each of the 8 arm tips: Receives the microwaves and turns them back into electricity for the launcher. • Sail-tilt station keeping (on each collector): Uses part of the light-pressure push to hold the collector in its slot (no propellant). • Clear sunlight lane: Collectors leave a moving gap so Earth and Mars see the whole Sun. • Collector factories (Mercury material): Build the belt up and replace aged panels.
E3	• Terminator orbit at 20 Earth radii + pointing (45 degree tilt): Keeps the mirrors in sunlight and aims their light at Earth or past it (the agent's action a4).

Support classification does not imply ordinary capability: launcher acceleration, power transfer, heat rejection, collector deployment and mirror pointing remain unverified and physically extraordinary.

Chain of cause and effect

Sunlight → **E2** thin-film PV field (electricity) → **supporting power-transfer equipment** (microwave transmitters and arm-tip rectennas) → **supporting launcher equipment** ejects finite packets from the **E1** asteroid resource → Earth recoils by physics → the changed orbit reduces direct sunlight → the **E3** mirror ring reflects additional sunlight into the simplified climate model. Supporting directional radiators and beam-spill control are separate safety assumptions; neither is included in an enabler label.

Key numbers at a glance

Quantity	Value
Required speed change of Earth (slow spiral)	11,658 m/s
Orbital energy Earth+Moon must gain	1.70×10^{33} J
Minimum launch energy (whole asteroid, slowest launch)	1.66×10^{35} J
Asteroid (reaction mass)	3.00×10^{22} kg = 0.408 Moon masses

Launch speed used by the hand-designed strategy	2,480 km/s = 0.83 % of light speed
Sunlight intercepted by the collector belt	1.15×10^{26} W (30 % of the Sun)
Electricity at the arms (Earth at 1 AU / at the target)	1.11×10^{25} W (weakest point)
Space-mirror capacity at the target	1.10 × today's sunlight on Earth
Analytic verdict	**FEASIBLE within this model (physically hypothetical scale)**
Historical pre-hold heuristic (high fidelity)	success: a = 2.6903 AU, e = 0.0072 after 3,600 y; surface 288.4 K

Numbers are labelled **C1, C2, …**. Each shows the formula, the values put in, the result and the function in `octoearth_rl.py` that does the same calculation in the simulation. Values marked *indicative* are NOT used by the simulation (they only help to judge an instrument).

Participant-guide crosswalk, reviewed 2026-10-27: the companion guide crosswalk and printable brief list the ten measured guide benchmarks, tolerances, judging criteria, enablers and open issues. `python octoearth_rl.py --check` runs all ten. Benchmark 8 is an isolated Sun–Earth two-body RK4 reference; benchmark 9 checks beam/cantilever equations only. Neither certifies the main launcher structure. Its segment/hinge SF ≥ 1.5 requirement remains **unverified** (problem P10 open); other appendix thresholds absent from the supplied PDFs are not guessed.

2. Physical constants and mission inputs

Constant	Value	Unit
Gravitational constant G	6.67430×10^{-11}	m^3 kg^-1 s^-2
Sun mass	1.98841×10^{30}	kg
Earth mass	5.97217×10^{24}	kg
Moon mass	7.34579×10^{22}	kg
Mars mass	6.41691×10^{23}	kg
Mercury mass	3.30100×10^{23}	kg
Venus mass	4.86731×10^{24}	kg
Jupiter mass	1.89852×10^{27}	kg
Mars radius	3.38950×10^{6}	m
Astronomical unit (AU)	1.49598×10^{11}	m
Year	3.15576×10^{7}	s
Earth radius	6.37100×10^{6}	m
Arm length	1.27420×10^{7}	m
Launch-ring radius	7.96375×10^{6}	m
Grab point height on the ring axis	7.96375×10^{6}	m
Moon distance	3.84400×10^{8}	m
Solar constant at 1 AU	1,361.00	W/m^2
Speed of light	2.99792×10^{8}	m/s

Mission input	Value
Target orbit radius a_t	2.7 AU (tolerance ±0.05 AU, e ≤ 0.05, i ≤ 5.0°)
Mission time limit T	10,000 years = 3.16×10^{11} s
Decision interval	100 years (100 decisions)
Climate target	288 K ± 5 K

The full list of all parameters with their values, units and meaning is in the appendix.

3. Mission requirement: how much push, energy and time

C1 **Mass that has to be moved**

Formula $m_{sys} = M_{Earth} + m_{asteroid} + M_{Moon}$

Values $5.9722\times10^{24} + 3.00\times10^{22} + 7.346\times10^{22}$ kg

Result **6.0756×10^{24} kg**

code: feasibility()

C2 **Gravitational parameter of the Sun-Earth system**

Formula $\mu = G\cdot M_{Sun} + G\cdot m_{sys}$

Values $1.32712\times10^{20} + 4.055\times10^{14}$ m³/s²

Result **1.32713×10^{20} m³/s²**

code: feasibility()

C3 **Circular orbital speed today (1 AU)**

Formula $v_0 = \sqrt{(\mu / a_0)}$

Values $\sqrt{(1.327\times10^{20} / 1.4960\times10^{11})}$

Result **29,784.7 m/s**

code: feasibility()

C4 **Circular orbital speed at the target (2.7 AU)**

Formula $v_t = \sqrt{(\mu / a_t)}$

Values $\sqrt{(1.327\times10^{20} / 4.0391\times10^{11})}$

Result **18,126.4 m/s**

code: feasibility()

C5 **Required speed change (slow outward spiral)**

Formula $\Delta v = v_0 - v_t$

Values 29,784.7 − 18,126.4

Result **11,658 m/s**

Meaning A slow spiral (small pushes along the motion over thousands of years) needs exactly the difference of the two circular speeds.

code: feasibility() dv_spiral

C6 **Reference: Hohmann transfer (two short burns)**

Formula $a_{tr} = (a_0+a_t)/2;\ \Delta v = |\sqrt{(\mu(2/a_0-1/a_{tr}))} - v_0| + |v_t - \sqrt{(\mu(2/a_t-1/a_{tr}))}|$

Values a_{tr} = 1.850 AU

Result **10,997 m/s**

Meaning Cheaper in speed change, but would need two enormous short pushes; the launcher can only push gradually, so the spiral value is used.

code: feasibility() dv_hoh

C7 **Orbital energy Earth + Moon (+asteroid) must gain**

Formula $\Delta E = G\ M_{Sun}\ m_{sys}\ (1/(2a_0) - 1/(2a_t))$

Values $1.3271\times10^{20} \times 6.076\times10^{24} \times (1/2.992\times10^{11} - 1/8.078\times10^{11})$

Result **1.70×10^{33} J**

code: feasibility() dE_orb

C8 Orbital angular momentum change

Formula $\Delta L = m_{sys} (\sqrt{(\mu\ a_t)} - \sqrt{(\mu\ a_0)})$

Result **1.74×10^{40} kg m²/s**

code: feasibility() dL_orb

C9 Average power just for the orbital energy

Formula $P = \Delta E / T$

Values 1.70×10^{33} J / 3.16×10^{11} s

Result **5.38×10^{21} W = 30,982 × all sunlight Earth intercepts**

code: feasibility()

C10 Escape speed from Earth + docked asteroid

Formula $v_{esc} = \sqrt{(2\ G\ (M_{Earth}+m_{ast}) / R_{Earth})}$

Values $\sqrt{(2 \times 6.6743\times10^{-11} \times 6.002\times10^{24} / 6.371\times10^{6})}$

Result **11,214 m/s**

Meaning A packet launched slower than this falls back; faster packets keep $v_\infty = \sqrt{(u^2 - v_{esc}^2)}$.

code: feasibility() vesc / launch_vinf()

C11 Rocket equation: mass ratio if the whole asteroid is thrown

Formula $MR = m_{sys} / (m_{sys} - m_{ast})$

Values 6.0756×10^{24} / 6.0456×10^{24}

Result **1.004962 (ln MR = 0.004950)**

code: feasibility()

C12 Speed change possible at the maximum launch speed

Formula $\Delta v_{cap} = v_{\infty,max} \cdot \ln MR$, $v_{\infty,max} = \sqrt{(u_{max}^2 - v_{esc}^2)}$

Values $5.0000\times10^{6} \times 0.004950$

Result **24,750 m/s (needed 11,658 m/s)**

code: feasibility() dv_cap_mass

C13 Reaction mass needed at the maximum launch speed

Formula $m = m_{sys} (1 - e^{-\Delta v / v_{\infty,max}})$

Result **1.41×10^{22} kg = 0.193 Moon masses**

code: feasibility() mass_needed_umax

C14 Slowest launch speed that works with the whole asteroid (minimum energy)

Formula $v_\infty = \Delta v / \ln MR$; $u = \sqrt{(v_\infty^2 + v_{esc}^2)}$

Values v_∞ = 11,658 / 0.004950 = 2.3552×10^{6} m/s

Result **u = 2.3552×10^{6} m/s**

code: feasibility() u_req

C15 Minimum launch energy

Formula $E_{min} = ½\ m_{ast}\ u^2 / \eta_{launch}$

Values $0.5 \times 3.00\times10^{22} \times (2.355\times10^{6})^2 / 0.5$

Result **1.66×10^{35} J (average 5.27×10^{23} W over the mission)**

code: feasibility() e_draw_min

C16 Shortest possible transfer time (maximum packet rate and speed)

Formula $\Delta v \text{ per year} = n \cdot m_{pk} \cdot v_{\infty,max} / m_{sys}$; $t_{min} = \Delta v / (\Delta v \text{ per year})$

Values $8 \times 1.00\times10^{18} \times 5.000\times10^{6} / 6.076\times10^{24} = 6.584$ m/s per year

Result **1,771 years**

code: feasibility() years_min

C17 Mass + time limits together

Formula $v_\infty = \max(\Delta v/\ln MR, \Delta v\, m_{sys}/(T\, n\, m_{pk}))$; $m = m_{sys}(1-e^{-\Delta v/v_\infty})$; $E = \tfrac{1}{2}\, m\, u^2/\eta$

Result **u = 2.3552×10^6 m/s, m = 3.00×10^22 kg, E = 1.66×10^35 J**

code: feasibility() combined

C18 Launcher peak power (maximum packet rate and speed)

Formula $P_{peak} = n \cdot m_{pk} \cdot \tfrac{1}{2}\, u_{max}^2 / \eta / \text{year}$

Values $8 \times 1.00\times10^{18} \times 0.5 \times (5.00\times10^{6})^2 / 0.5 / 3.1558\times10^{7}$

Result **6.34×10^24 W**

code: feasibility() p_peak

C19 Energy available over the mission

Formula $E_{avail} = \text{reserve} + P_{elec,min} \cdot T$

Values $3.00\times10^{35} + 1.11\times10^{25} \times 3.16\times10^{11}$

Result **3.82×10^36 J**

code: feasibility() e_avail

Check	Result	Detail
mass	**OK**	needs launch speed 2.355e+06 m/s with ALL 3.00e+22 kg; max allowed 5.000e+06 m/s (mass-limited delta-v capacity 2.475e+04 m/s vs required 1.166e+04 m/s)
energy	**OK**	minimum energy drawn 1.664e+35 J vs available 3.817e+36 J (reserve 3.000e+35 J + collectors 1.114e+25 W x 10,000 y)
rate	**OK**	at max packet rate and speed the transfer needs >= 1,771 years (mission 10,000 y)
combined	**OK**	slowest speed meeting mass AND time limits 2.355e+06 m/s -> needs 3.000e+22 kg and 1.664e+35 J

Verdict: FEASIBLE within this model (physically hypothetical scale). The realistic preset (Ceres-mass asteroid, 30 km/s launcher, Earth's own sunlight as energy) is infeasible: verdict *INFEASIBLE with these parameters (failed: mass, energy, rate, combined; mass-limited delta-v reaches 3.71e-04 of the requirement)*.

4. Enabler E1 — finite asteroid access; reaction-mass and launcher-support calculations

Selected enabler: Access to a finite 3e22 kg asteroid
A stated asteroid mass reservoir is available as reaction mass; the simulated launch packets are finite and Earth recoils when they are ejected. This is one resource-access assumption, not a launcher-system bundle.
Why selected: Without reaction mass the modeled recoil mechanism cannot transfer momentum to Earth.
Gap to reality: No asteroid is actually captured or docked; access to and placement of a 3e22 kg body is far beyond demonstrated capability.
Modelled: Finite mass remaining, packet ejection, recoil, packet fates and mass/energy/momentum accounting. Hardware, capture and launch shock are not simulated; Earth-damage question P6 remains open.

Selected enabler E1.1 - Finite asteroid-access assumption and reaction-mass reservoir

Selected assumption: access to a finite 3e22 kg asteroid. The simulation tracks that configured reservoir as reaction mass and debits ejected packets. It does **not** simulate capturing, docking or placing an asteroid of this size. Earth can only change its speed by throwing mass the other way (momentum conservation).

C20 Reservoir size compared with the Moon and Earth

Formula	m_{ast} / M_{Moon}, m_{ast} / M_{Earth}
Values	$3.00\times10^{22} / 7.346\times10^{22}$; $3.00\times10^{22} / 5.972\times10^{24}$
Result	**0.408 Moon masses; 0.502 % of Earth**

code: Config.asteroid_mass_kg

C21 Surface gravity with the asteroid attached (indicative)

Formula	$g = G\,(M_{Earth} + m_{ast}) / R_{Earth}^2$
Result	**9.8696 m/s² (today 9.8203 m/s², +0.50 %)**
Meaning	The model treats Earth + asteroid as one body (its gravity counts in every orbit calculation).

code: indicative

C22 Number of full packets in the reservoir

Formula	$N = m_{ast} / m_{pk,max}$
Values	$3.00\times10^{22} / 1.00\times10^{18}$
Result	**30,000 packets = 3,750 years at the maximum rate**

C23 Tested mission: reservoir used

Formula	m_{used} / m_{ast}
Values	$2.86\times10^{22} / 3.00\times10^{22}$
Result	**95.2 % in 28,560 packets**

code: summary() mass_used_frac

Supporting system E1.2 - Launch ring with 8 arms (mass driver)

Use: accelerates each packet to the commanded speed u (between 20.0 and 5,000 km/s) in the commanded direction; 8 launches per year. All 8 arms throw ONE packet together (group throw), so each arm carries 1/8 of the packet and of the energy.

C24 **Launch speed chosen by the hand-designed strategy**

Formula $v_\infty = \max(\Delta v / \ln(m_{sys}/(m_{sys}-0.95\ m_{ast})),\ \Delta v\ m_{sys} / (0.85\ T\ n\ m_{pk}))$; $u = \sqrt{(v_\infty{}^2+v_{esc}{}^2)}$

Values $\max(2.479\times10^{6},\ 1.042\times10^{6})$ m/s

Result **$u = 2.4795\times10^{6}$ m/s = 2,480 km/s (0.83 % of c)**

Meaning Lowest speed that uses 95 % of the asteroid and still finishes within 85 % of the mission time.

code: heuristic_speed()

One full packet at the strategy speed (u = 2,480 km/s)

C25 **Kinetic energy given to packet + Earth (two-body, reduced mass)**

Formula $\mu_{red} = m_{pk}(M-m_{pk})/M$; $E_{kin} = \tfrac{1}{2}\ \mu_{red}\ u^2$

Values $\mu_{red} = 1.00000\times10^{18}$ kg; $0.5 \times 1.000\times10^{18} \times (2.480\times10^{6})^2$

Result **3.07×10^{30} J**

code: eject_core()

C26 **Energy drawn from the power supply**

Formula $E_{draw} = E_{kin} / \eta_{launch}$

Values $3.07\times10^{30} / 0.5$

Result **6.15×10^{30} J per packet, 7.68×10^{29} J per arm; 6.15×10^{6} × a Chicxulub impact (~10^{24} J)**

code: eject_core()

C27 **Packet speed after leaving Earth's and the Moon's gravity**

Formula $v_\infty = \sqrt{(u^2 - 2GM/R_{Earth} - 2GM_{Moon}/d_{Moon})}$

Values $\sqrt{((2.480\times10^{6})^2 - 1.258\times10^{8} - 25{,}509)}$

Result **2.47949×10^{6} m/s**

code: eject_core() (fast model)

C28 **Recoil: speed change of Earth + Moon per packet**

Formula $\Delta v_E = m_{pk}\ v_\infty / (M_{Earth}+m_{ast}+M_{Moon})$

Values $1.00\times10^{18} \times 2.479\times10^{6} / 6.0756\times10^{24}$

Result **0.408 m/s (impulse 2.48×10^{24} N s)**

code: _fast_eject() / ledger LG_IMPULSE_TOTAL

C29 **Full-throttle push per year and per decision**

Formula $\Delta v_{yr} = n \cdot \Delta v_E$

Values 8×0.408

Result **3.26 m/s per year, 326 m/s per 100-year decision; Δv needs 3,571 years of full thrust**

code: policy_heuristic() dv_max

C30 **Average launcher power at full throttle**

Formula $P = n \cdot E_{draw} / year$

Values $8 \times 6.15\times10^{30} / 3.1558\times10^{7}$

Result **1.56×10^{24} W (1.95×10^{23} W per arm)**

One full packet at the maximum speed (u = 5,000 km/s)

C31 **Kinetic energy given to packet + Earth (two-body, reduced mass)**

Formula $\mu_{red} = m_{pk}(M-m_{pk})/M$; $E_{kin} = ½\ \mu_{red}\ u^2$

Values $\mu_{red} = 1.00000\times10^{18}$ kg; $0.5 \times 1.000\times10^{18} \times (5.000\times10^{6})^2$

Result **1.25×10^{31} J**

code: eject_core()

C32 **Energy drawn from the power supply**

Formula $E_{draw} = E_{kin} / \eta_{launch}$

Values $1.25\times10^{31} / 0.5$

Result **2.50×10^{31} J per packet, 3.12×10^{30} J per arm; 2.50×10^{7} × a Chicxulub impact (~10^{24} J)**

code: eject_core()

C33 **Packet speed after leaving Earth's and the Moon's gravity**

Formula $v_\infty = \sqrt{(u^2 - 2GM/R_{Earth} - 2GM_{Moon}/d_{Moon})}$

Values $\sqrt{((5.000\times10^{6})^2 - 1.258\times10^{8} - 25{,}509)}$

Result **4.99999×10^{6} m/s**

code: eject_core() (fast model)

C34 **Recoil: speed change of Earth + Moon per packet**

Formula $\Delta v_E = m_{pk}\ v_\infty / (M_{Earth}+m_{ast}+M_{Moon})$

Values $1.00\times10^{18} \times 5.000\times10^{6} / 6.0756\times10^{24}$

Result **0.823 m/s (impulse 5.00×10^{24} N s)**

code: _fast_eject() / ledger LG_IMPULSE_TOTAL

C35 **Full-throttle push per year and per decision**

Formula $\Delta v_{yr} = n \cdot \Delta v_E$

Values 8×0.823

Result **6.58 m/s per year, 658 m/s per 100-year decision; Δv needs 1,771 years of full thrust**

code: policy_heuristic() dv_max

C36 **Average launcher power at full throttle**

Formula $P = n \cdot E_{draw} / year$

Values $8 \times 2.50\times10^{31} / 3.1558\times10^{7}$

Result **6.34×10^{24} W (7.92×10^{23} W per arm)**

C37 **Time between launches**

Formula $\Delta t = 1\ \text{year} / n$

Values 365.25 d / 8

Result **45.66 days**

code: fast_interval()

C38 **Mass per arm in a group throw**

Formula $m_{arm} = m_{pk} / 8$

Values 1.00×10^{18} / 8

Result **1.25×10^{17} kg**

code: N_ARMS = 8

C39 **Tested mission: total impulse and speed change delivered**

Formula $\Delta v = J_{total} / M_{Earth+Moon+ast}$

Values 7.08×10^{28} N s / 6.0756×10^{24} kg

Result **11,655 m/s (required by the spiral estimate: 11,658 m/s)**

Meaning Compare with the slow-spiral estimate C5: the hand-designed strategy pushes almost purely along Earth's motion, so the delivered speed change is close to the ideal value.

code: ledger LG_IMPULSE_TOTAL

C40 **Tested mission: packet kinetic energy and launcher draw**

Formula $\eta = E_{kin} / E_{draw}$

Values 8.78×10^{34} / 1.76×10^{35}

Result **0.5000 (= launch_efficiency 0.5)**

code: ledger LG_LAUNCH_KE, LG_E_DRAWN

C41 **Tested mission: average launcher power**

Formula $P = E_{draw} / t$

Values 1.76×10^{35} / (3,600 y × 3.1558×10^{7} s)

Result **1.55×10^{24} W**

Launcher geometry: asteroid on the ring axis, arms 2 Earth radii long (update p36)

The 8 arm roots sit on a ring around Earth (plane perpendicular to Earth's orbit normal). The docked asteroid sits on the ring's **axis**; its Earth-facing side rests on the grab point, so every arm root is exactly the same distance from it and all 8 arms reach it equally. The docking clamps lower the asteroid as it shrinks, so the grab point never moves. Each arm is 2 Earth radii long.

C42 **Distance from every arm root to the grab point**

Formula $d = \sqrt{(\rho_{ring}^2 + z_{grab}^2)} \le L_{arm}$

Values $\sqrt{(1.25^2 + 1.25^2)}$ R_E; L_{arm} = 2.0 R_E = 12,742 km

Result **1.7678 R_E = 11,262 km for all 8 roots (spread 0e+00); reach to spare 1,480 km**

code: launcher_geometry()

C43 **Asteroid size and position**

Formula $r_{ast} = (3\ m_{ast} / (4\pi\rho))^{1/3}$; $z_{centre} = z_{grab} + r_{ast}$

Values ρ = 2000 kg/m³ (assumed), $m_{ast} = 3.00\times10^{22}$ kg

Result **r_{ast} = 1,530 km; centre 1.490 R_E from Earth's centre, 1.945 R_E from every root (≤ arm length); 1,593 km above the surface**

code: launcher_geometry()

C44 **Stroke of the joint throw (all 8 arms keep hold)**

Formula largest s with $|G + s\cdot e - root_k| \le L_{arm}$ for all k (worst direction e in the ring plane)

Values bisection over directions

Result **s = 1,983 km (0.311 R_E; up to 0.330 R_E between two arms); release point 1.288 R_E from Earth's centre**

code: launcher_geometry()

C45 **Acceleration the arms must give a packet over that stroke**

Formula $a = u^2 / (2\ s)$; $t = 2\ s / u$

Values u = 2,480 km/s (strategy) and 5,000 km/s (maximum); s = 1,983 km

Result **1.55×10^6 m/s² (1.58×10^5 g) for 1.60 s; at the maximum speed 6.30×10^6 m/s²**

Meaning Far beyond any material; this is a gap in the separately classified launcher support system.

code: launcher_geometry()

C46 **Load on the docking clamps (asteroid's weight)**

Formula $F = m_{ast}\ G\ M_{Earth} / z_{centre}^2$

Values 3.00×10^{22} × 4.423 m/s²

Result **1.33×10^{23} N**

code: launcher_geometry()

C47 **What the physics simplifies (and why it is allowed)**

Formula $\Delta E/E = (GM/R)(1 - R/r_{release}) / (u^2/2)$; COM shift = $m_{ast}\ z_{centre}/(M+m_{ast})$

Values escape energy from Earth's surface instead of the release point; asteroid merged with Earth

Result **surface launch needs $+4.57\times10^{-6}$ more energy (conservative); centre of mass off by 47 km; tidal error on the Moon 9.41×10^{-6} of Earth's pull**

code: launcher_geometry(), check 21

Not modelled / not validated: launcher structure and materials, packet acceleration inside the arms, and each launch's effect on Earth (shock, atmosphere loss and heating). These are unresolved supporting-system hazards, not granted by E1. Open problem P6.

Supporting system E1.3 - Arm power conditioning and energy reserve

Use: the arms take electricity from the rectennas (E2) first; if a launch needs more than the beams deliver in that interval, the rest comes from a finite reserve. Unused beam energy is not sent (the beams are throttled to demand); nothing is created from nothing.

C48 **Reserve size compared with the minimum mission energy**

Formula $\text{reserve} / E_{min}$

Values 3.00×10^{35} / 1.66×10^{35}

Result **1.803**

code: Config.energy_budget_j

C49 **Energy offered by the beams in one decision interval (Earth at 1 AU)**

Formula $E_{beam} = P_{elec} \cdot \Delta t_{decision}$

Values 1.29×10^{25} W × 100 y

Result **4.07×10^{34} J**

code: _PowerSupply._with_supply()

C50 **Tested mission: where the launch energy came from**

Formula $E_{draw} = E_{beams} + E_{reserve}$

Values 1.76×10^{35} + 0

Result **1.76×10^{35} J; beams used 12.8 % of what they could offer**

code: summary() energy_from_collectors_J

Supporting system E1.4 - Directional arm radiators

Use: the launcher loses (1−η) = 50 % of the energy it draws as heat, and the rectennas lose ($1-\eta_r$) = 15 % of the microwaves they catch. The radiators send this heat to space, away from Earth; only the share arm_heat_to_earth_frac = 1.00×10^{-9} reaches Earth.

C51 **Tested mission: arm waste heat**

Formula $Q = E_{\text{rectenna heat}} + E_{\text{launcher heat}}$

Values 3.10×10^{34} + 8.78×10^{34} J

Result **1.19×10^{35} J = 1.05×10^{24} W on average = 6.02×10^{6} × the sunlight Earth intercepts**

code: settle_interval()

C52 **Radiator area needed at 1,000 K (indicative, black radiator, one side)**

Formula $A = Q / (\sigma\ T\&sup4;)$

Values 1.05×10^{24} / (5.670×10^{-8} × 1000&sup4;)

Result **1.84×10^{19} m² in total, 2.30×10^{18} m² per arm = 18,074 × Earth's disc**

code: summary() radiator_area_1000K_m2

C53 **Heat that still reaches Earth**

Formula $q_{Earth} = f_{arm} \cdot Q + f_{spill} \cdot E_{spill}$

Values 1.00×10^{-9} × 1.19×10^{35} + 1.00×10^{-9} × 5.38×10^{33}

Result **1.24×10^{26} J = 1.09×10^{15} W = 0.00630 × Earth's sunlight**

code: _climate_step() q_w

Without the directional radiators a geometric leak of ~8 % would put 8.36×10^{22} W (4.82×10^{5} × Earth's sunlight) on Earth: the oceans would boil. Directional radiators are separate supporting hardware, assumed here and not physically designed.

Mars safety limits (control software, not an enabler)

Earth's growing orbit crosses Mars's orbit. The safety shield forecasts the Earth-Mars distance and overrides the agent (coast or full thrust) when a pass would come too close. The safe distance is the largest of three limits:

C54 **Tide limit: Mars's tide on Earth ≤ 1 % of the Moon's**

Formula $d_{tide} = d_{Moon} (GM_{Mars} / (GM_{Moon} \cdot k_{tide}))^{1/3}$

Values $3.844\times10^{8} \times (4.283\times10^{13} / (4.903\times10^{12} \times 0.01))^{1/3}$

Result **3.67×10^{6} km**

code: mars_safety_limits()

C55 **Moon-orbit limit: Mars's pull on the Moon's orbit ≤ 0.1 % of Earth's**

Formula $d_{moon} = d_{Moon} (2\ GM_{Mars} / (GM_{Earth}\ k_{moon}))^{1/3}$

Values $k_{moon} = 0.001$

Result **2.30×10^{6} km**

code: mars_safety_limits()

C56 **Safe distance and shield caution distance**

Formula $d_{safe} = \max(d_{tide}, d_{moon}, d_{min})$; $d_{caution} = k_c\ d_{safe}$

Values $\max(3.67\times10^{6}, 2.30\times10^{6}, 2.00\times10^{6})$ km; $k_c = 2.0$

Result **$d_{safe} = 3.67\times10^{6}$ km = 0.0246 AU (governed by: tide on Earth); caution 0.0491 AU**

code: mars_safety_limits()

Tested mission: closest Earth-Mars pass 0.0502 AU (≥ 0.0246 AU: safe), largest tide ratio 0.00117, 272 shield overrides. Mars's own orbit changed by Δa = 0.0340 AU, Δe = -9.08×10^{-4}.

Mercury, Venus and Jupiter (update p35; Saturn and beyond not modelled)

Mercury, Venus and Jupiter start on their real (J2000) orbits and are integrated in both models: they pull on Earth, the Moon (high-fidelity model), Mars and the fragments, and Earth pulls back (two-way gravity). The same three safety rules as for Mars are applied to each planet; because the tide limits scale with GM, the safe distance scales with $GM^{1/3}$. Coming closer ends the mission (failure planet_unsafe_approach). The shield's forecast still uses only Sun + Earth + Mars.

C57 **Safe distance from Mercury**

Formula $d_{safe} = \max(d_{Moon}(GM/(GM_{Moon}k_{tide}))^{1/3}, d_{Moon}(2GM/(GM_{Earth}k_{moon}))^{1/3}, d_{min})$

Values GM = 2.203×10^{13} m³/s²; tide 2.94×10^{6} km, Moon-orbit 1.84×10^{6} km, minimum 2.00×10^{6} km

Result **2.94×10^{6} km = 0.0197 AU (governed by: tide on Earth)**

code: planet_safety_limits()

C58 **Safe distance from Venus**

Formula $d_{safe} = \max(d_{Moon}(GM/(GM_{Moon}k_{tide}))^{1/3}, d_{Moon}(2GM/(GM_{Earth}k_{moon}))^{1/3}, d_{min})$

Values GM = 3.249×10^{14} m³/s²; tide 7.22×10^{6} km, Moon-orbit 4.52×10^{6} km, minimum 2.00×10^{6} km

Result **7.22×10^{6} km = 0.0483 AU (governed by: tide on Earth)**

code: planet_safety_limits()

C59 **Safe distance from Jupiter**

Formula $d_{safe} = \max(d_{Moon}(GM/(GM_{Moon}k_{tide}))^{1/3}, d_{Moon}(2GM/(GM_{Earth}k_{moon}))^{1/3}, d_{min})$

Values GM = 1.267×10^{17} m³/s²; tide 5.28×10^{7} km, Moon-orbit 3.31×10^{7} km, minimum 2.00×10^{6} km

Result **5.28×10^{7} km = 0.3526 AU (governed by: tide on Earth)**

code: planet_safety_limits()

C60 **Pull of Mercury on Earth (two-way gravity in both models)**

Formula $a = GM / d^2$ (and Earth pulls back with $G\,M_{Earth} / d^2$)

Values GM = 2.203×10^{13} m³/s²

Result **at the closest approach of the tested mission (0.550 AU): 3.25×10^{-9} m/s²; at 2.7 AU with Mercury at its nearest (2.31 AU): 1.84×10^{-10} m/s² = 2.26×10^{-7} × the Sun's pull there**

code: _planet_kicks() (fast) / nb_accel() (N-body)

C61 **Pull of Venus on Earth (two-way gravity in both models)**

Formula $a = GM / d^2$ (and Earth pulls back with $G\,M_{Earth} / d^2$)

Values GM = 3.249×10^{14} m³/s²

Result **at the closest approach of the tested mission (0.267 AU): 2.03×10^{-7} m/s²; at 2.7 AU with Venus at its nearest (1.98 AU): 3.71×10^{-9} m/s² = 4.57×10^{-6} × the Sun's pull there**

code: _planet_kicks() (fast) / nb_accel() (N-body)

C62 **Pull of Jupiter on Earth (two-way gravity in both models)**

Formula $a = GM / d^2$ (and Earth pulls back with $G\,M_{Earth} / d^2$)

Values GM = 1.267×10^{17} m³/s²

Result **at the closest approach of the tested mission (2.325 AU): 1.05×10^{-6} m/s²; at 2.7 AU with Jupiter at its nearest (2.50 AU): 9.04×10^{-7} m/s² = 0.00111 × the Sun's pull there**

code: _planet_kicks() (fast) / nb_accel() (N-body)

planet	closest approach (tested mission, N-body)	safe distance	verdict	orbit change Δa
Mercury	0.550 AU (year 0)	0.0197 AU	safe	+2.1e-06 AU
Venus	0.267 AU (year 8)	0.0483 AU	safe	-9.3e-06 AU
Jupiter	2.325 AU (year 3567)	0.3526 AU	safe	+1.5e-04 AU

Jupiter, 318 Earth masses, dominates the effect on Earth's final orbit; the 8:3 mean-motion resonance with Jupiter lies at about 2.705 AU, very close to the 2.7 AU target, so near the target Jupiter's periodic pull slowly changes Earth's eccentricity. Over the simulated mission this stays within the tolerance; over millions of years it is NOT checked here (long-term stability would need a dedicated study).

5. Enabler E2 — thin-film photovoltaic collector field; transfer-support calculations

Selected enabler: Thin-film photovoltaic collector field
Five million thin-film collectors at 0.5 AU intercept the configured 30 % of the Sun's output and convert sunlight into electricity. This is one power-generation enabler; microwave transmitters, rectennas and factories are supporting components, not bundled enablers.
Why selected: The modeled mission needs a physical energy source; sunlight is traced through panel conversion and the downstream ledger.
Gap to reality: A Dyson-swarm-scale photovoltaic field (~1e13 x all solar panels ever built); collector area and mass are far beyond industry. The power-transfer chain separately assumes about 1e24 W over 0.5-3 AU, pointing near 1e-12 rad, and rectenna beam intensity around 2e8 times sunlight.
Modelled: Panel temperature and efficiency, degradation and replacement, build-up, manufacturing energy and mass, Goubau diffraction capture, Sun blocking, every loss in the energy ledger, clear lanes, the orbits of 12+12 followed collectors with / without station keeping (beam recoil, planets, light pressure).

Selected enabler E2.1 - Thin-film photovoltaic collector field

Use: intercept sunlight and turn it into electricity. The belt is a band around the Sun at 0.5 AU that covers the fraction f = 0.3 of a full sphere, so it intercepts exactly that fraction of all sunlight.

C63 Belt area

Formula $A = f \cdot 4\pi R^2$
Values $0.3 \times 4\pi \times (7.480\times10^{10})^2$
Result **2.11×10^{22} m² = 4.14×10^{7} × Earth's surface**

code: collector_design()

C64 Band half-width (latitude seen from the Sun)

Formula $\beta_{band} = \arcsin f$
Values $\arcsin 0.3$
Result **17.46°**

code: collector_design()

C65 Size of one collector

Formula $A_1 = A / N$; side $= \sqrt{A_1}$
Values $2.11\times10^{22} / 5.00\times10^{6}$
Result **4.22×10^{15} m², about 64,950 km × 64,950 km**

code: collector_design()

C66 Mass of the belt and of one collector

Formula $m = A \cdot \sigma_{areal}$
Values $2.11\times10^{22} \times 0.1$ kg/m²
Result **2.11×10^{21} kg (0.0287 Moon masses); one collector 4.22×10^{14} kg**

code: collector_design()

C67 **Sunlight at the belt**

Formula $S = S_{1AU} / r^2$

Values $1361.0 / 0.5^2$

Result **5,444.0 W/m²**

code: collector_operating()

C68 **Sunlight intercepted (whole belt built)**

Formula $L_{cap} = f \cdot L_{Sun}$

Values $0.3 \times 3.828 \times 10^{26}$

Result **1.15×10^{26} W**

code: collector_power()

C69 **Panel temperature (energy balance, both faces radiate)**

Formula $T = [(a\ S - \eta(T)\ S) / (2\ \varepsilon\ \sigma)]^{1/4}$, $\eta(T) = \eta_{25°C}(1 - k\ (T - 298.15\ K))$

Values $a = 0.9$, $\varepsilon = 0.85$, $\eta_{25°C} = 0.25$, $k = 0.002/K$ (solved by iteration)

Result **T = 450.0 K, η(T) = 17.41 % (×0.696); idle panel (no electricity) 474.8 K**

code: collector_operating()

C70 **Ageing of the fleet (panels replaced at end of life)**

Formula $f_{age} = 1 - d \cdot L_{life} / 2$

Values $1 - 0.002 \times 50 / 2$

Result **0.950**

code: collector_operating()

C71 **Operating solar-cell efficiency**

Formula $\eta_{op} = \eta(T) \cdot f_{age}$

Values 0.1741×0.950

Result **16.54 %**

code: collector_operating()

C72 **Panel electricity (whole belt, clear lanes subtracted)**

Formula $P_{pv} = f\ L_{Sun}\ (1 - lanes)\ \eta_{op}$

Values $1.15 \times 10^{26} \times (1 - 5.11 \times 10^{-5}) \times 0.1654$

Result **1.90×10^{25} W; one collector 3.80×10^{18} W**

code: collector_power()

C73 **Light-pressure ratio of a collector (push of sunlight / pull of the Sun)**

Formula $\beta = L_{Sun} / (4\pi\ c\ GM_{Sun}\ \sigma_{areal})$

Values $3.828 \times 10^{26} / (4\pi \times 2.998 \times 10^{8} \times 1.327 \times 10^{20} \times 0.1)$

Result **0.00766 (0.77 %)**

code: collector_design()

C74 **Orbital period of a collector**

Formula $P = r^{3/2} / \sqrt{(1 - \beta)}$ years

Values $0.5^{1.5} / \sqrt{(1 - 0.0077)}$

Result **0.3549 years = 129.6 days**

code: collector_design()

C75 **Gravity of the whole belt on Earth (vs the Sun's pull)**

Formula $a = \Sigma\, G\, m_i\, d_i/|d_i|^3$ over 6,480 belt points

Result **1.28×10^{-9} of the Sun's pull at 1 AU, 1.09×10^{-9} at 2.7 AU (negligible)**

code: belt_pull_on_earth()

Supporting system E2.2 - Microwave phased-array transmitters

Use: each collector turns its electricity into a 5.8 GHz beam with a 30 km phased array and points it at the arms. Efficiency 80 %; the beams are throttled to what the launcher needs.

C76 **Wavelength**

Formula $\lambda = c / f$

Values 2.99792×10^{8} / 5.80×10^{9}

Result **5.169 cm**

code: beam_capture_efficiency()

C77 **Microwave power sent (whole belt, full output)**

Formula $P_{mw} = P_{pv}\, \eta_{mw}$

Values $1.90\times10^{25} \times 0.8$

Result **1.52×10^{25} W; per transmitter 3.04×10^{18} W**

code: collector_power()

C78 **Intensity on a transmitter aperture at full output (indicative)**

Formula $I = (P_{mw}/N) / (\pi\, (D_t/2)^2)$

Values 3.04×10^{18} / 7.07×10^{8} m²

Result **4.30×10^{9} W/m² = 3.16×10^{6} × sunlight at Earth**

code: indicative

C79 **Beam capture at 0.50 AU (Goubau relation)**

Formula $\tau = D_t\, D_r / (\lambda\, d)$; $\eta_{cap} = 1 - e^{-\tau^2}$

Values $\tau = 30{,}000 \times 1.00\times10^{6} / (0.05169 \times 7.480\times10^{10}) = 7.759$

Result **η_{cap} = 100.00 %**

code: beam_capture_efficiency()

C80 **Central beam spot diameter at 0.50 AU (indicative, Airy)**

Formula $D_{spot} \approx 2.44\, \lambda\, d / D_t$

Values $2.44 \times 0.05169 \times 7.480\times10^{10}$ / 30,000

Result **314 km (rectenna 1000 km)**

code: indicative

C81 **Beam capture at 1.50 AU (Goubau relation)**

Formula $\tau = D_t\, D_r / (\lambda\, d)$; $\eta_{cap} = 1 - e^{-\tau^2}$

Values $\tau = 30{,}000 \times 1.00\times10^{6} / (0.05169 \times 2.244\times10^{11}) = 2.587$

Result **η_{cap} = 99.88 %**

code: beam_capture_efficiency()

C82 **Central beam spot diameter at 1.50 AU (indicative, Airy)**

Formula $D_{spot} \approx 2.44\ \lambda\ d\ /\ D_t$

Values $2.44 \times 0.05169 \times 2.244{\times}10^{11}\ /\ 30{,}000$

Result **943 km (rectenna 1000 km)**

code: indicative

C83 **Average capture over the whole belt, Earth at 1 AU / at the target**

Formula $\bar{\eta}_{cap}$ = mean over 6,480 belt points (beams within 2 R_{Sun} of the Sun blocked)

Values blocked: 0.1 % / 0.1 % of the belt

Result **99.88 % / 86.30 %**

code: collector_power()

C84 **Pointing accuracy needed (indicative)**

Formula $\delta\theta \approx (D_r/2)\ /\ d_{max}$

Values $5.00{\times}10^{5}$ m / $4.787{\times}10^{11}$ m

Result **$1.04{\times}10^{-6}$ rad**

code: indicative

C85 **Tested mission: transmitter waste heat and beam spill**

Formula $Q_{tx} = E_{el}(1-\eta_{mw})$; $E_{spill} = E_{mw}(1-\eta_{cap})$

Values $2.65{\times}10^{35} \times 0.20$; $2.12{\times}10^{35} - 2.07{\times}10^{35}$

Result **Q_{tx} = $5.30{\times}10^{34}$ J; spill $5.38{\times}10^{33}$ J (2.54 %), of which $1.00{\times}10^{-9}$ reaches Earth**

code: settle_interval()

Supporting system E2.3 - Rectennas on the 8 arm tips

Use: receive the microwaves (only the arms receive power) and turn them back into electricity with 85 % efficiency. Diameter 1000 km each.

C86 **Area of one rectenna**

Formula $A_r = \pi\ (D_r/2)^2$

Values $\pi \times (5.00{\times}10^{5})^2$

Result **$7.85{\times}10^{11}$ m² (0.0493 × Earth's disc for all 8)**

code: collector_design()

C87 **Microwaves arriving (full output), Earth at 1 AU / at the target**

Formula $P_{in} = P_{mw}\ \bar{\eta}_{cap}$

Values $1.52{\times}10^{25} \times 0.9988$; $1.52{\times}10^{25} \times 0.8630$

Result **$1.52{\times}10^{25}$ W / $1.31{\times}10^{25}$ W**

code: collector_power()

C88 **Electricity delivered to the arms (full output)**

Formula $P_{elec} = P_{in}\ \eta_r$

Values $1.52{\times}10^{25} \times 0.85$

Result **$1.29{\times}10^{25}$ W at 1 AU, $1.11{\times}10^{25}$ W at 2.7 AU; at year 0 (only 30 % built) $3.87{\times}10^{24}$ W**

code: collector_power()

C89 **Supply vs the launcher's peak demand**

Formula $P_{elec,min} / P_{peak}$

Values $1.11\times10^{25} / 6.34\times10^{24}$

Result **1.76 ×**

code: feasibility()

C90 **Beam intensity on a rectenna at full output (Earth at 1 AU)**

Formula $I = P_{in} / (8\ A_r)$

Values $1.52\times10^{25} / (8 \times 7.85\times10^{11})$

Result **2.42×10^{12} W/m² = 1.77×10^{9} × sunlight**

code: collector_design()

C91 **Tested mission: average and peak beam power received**

Formula $\bar{P} = E_{captured} / t$

Values 2.07×10^{35} J / (3,600 y)

Result **mean 1.82×10^{24} W (2.27×10^{23} W per arm, 2.89×10^{11} W/m²); peak 1.83×10^{24} W**

code: summary() beam_mean_W

C92 **Tested mission: rectenna heat**

Formula $Q_r = E_{captured}\ (1 - \eta_r)$

Values $2.07\times10^{35} \times 0.15$

Result **3.10×10^{34} J (assumed rejected by separate directional radiator hardware)**

code: settle_interval()

C93 **Tested mission: push of the absorbed microwaves on Earth (indicative side effect)**

Formula $\Delta v = E_{in} / (c\ M)$

Result **114 m/s (counted in the report, not applied)**

code: summary() beam_push_dv_ms

Supporting system E2.4 - Sail-tilt station keeping

Use: each collector tilts slightly so the sunlight push corrects its orbit (no propellant). It may use up to collector_sk_authority = 0.5 of its light push. 12 collectors are simulated with it and the same 12 without it (RK4, half-day steps) along Earth's and Mars's simulated motion.

C94 **Sun's pull at the belt**

Formula $g = GM_{Sun} / R^2$

Values $1.3271\times10^{20} / (7.480\times10^{10})^2$

Result **0.0237 m/s²**

code: BeltKeeper

C95 **Largest control acceleration**

Formula $a_{ctrl} = k_{sk}\ \beta\ GM_{Sun} / R^2$

Values $0.5 \times 0.00766 \times 0.0237$

Result **9.08×10^{-5} m/s² = 0.00383 of the Sun's pull**

code: BeltKeeper.auth

C96 **Controller speed (PD gains)**

Formula $\omega_c = 4\ n,\ n = \sqrt{(GM_{Sun}(1-\beta)/R^3)}$

Values $n = 5.61\times10^{-7}$ rad/s

Result **$\omega_c = 2.24\times10^{-6}$ rad/s (time constant 5.2 days)**

code: BeltKeeper.wc

C97 **Beam recoil on a collector (full output)**

Formula $a_{rec} = (P_{mw}/N) / (c\ m_1)$

Values $3.04\times10^{18} / (2.998\times10^{8} \times 4.22\times10^{14})$

Result **2.40×10^{-5} m/s² = 0.00101 of the Sun's pull**

code: BeltKeeper.advance() a_rec

C98 **Pull of a neighbouring collector**

Formula $a = G\ m_1 / s^2$

Values $6.6743\times10^{-11} \times 4.22\times10^{14} / (6.495\times10^{7})^2$

Result **2.81×10^{-10} of the Sun's pull**

code: BeltKeeper.neighbour_ratio

Tested mission (3,600 years)	WITH station keeping	WITHOUT
largest distance from the slot	730 km	1.54×10^{8} km
... in collector spacings (64,950 km)	0.0112	2,371
orbit size change |Δa|	1,158 km	25,044 km
largest eccentricity	1.07×10^{-5}	0.109
control effort	96.8 m/s per year	0

Largest disturbances (fraction of the Sun's pull): beam recoil 1.39×10^{-4}, planets 2.48×10^{-6}, neighbours 2.81×10^{-10}; the controller can correct up to 0.00383.

Supporting system E2.5 - Clear sunlight lane

Use: the collectors leave a moving gap on the line Sun-Earth (and Sun-Mars) so that Earth sees the whole solar disc.

C99 **Share of the belt that must stay empty for Earth**

Formula $\text{lane} = \rho^2 / (4\ R^2\ f),\ \rho = R_{Sun}(r-R)/r + R_{Earth}\ R/r$

Values $r = 1$ AU

Result **1.84×10^{-5} (Mars: 3.27×10^{-5})**

code: lane_fraction()

C100 **Power given up for the lanes**

Formula $\Delta P = L_{cap} \times \text{lanes} \times \eta_{op}$

Values $1.15\times10^{26} \times 5.11\times10^{-5} \times 0.1654$

Result **9.70×10^{20} W of electricity**

code: collector_power()

Supporting system E2.6 - Collector factories (Mercury material)

Use: 30 % of the belt exists at year 0; factories finish it in 400 years and replace panels after 50 years (95 % recycled). Manufacturing energy comes from the collectors' own output.

C101 **Built share over time**

Formula $b(t) = b_0 + (1 - b_0)\ t / T_{build}$ (max 1)

Values $b_0 = 0.3$, $T_{build} = 400$ y

Result **b(100 y) = 47.5 %, b(200 y) = 65.0 %, complete at 400 y**

code: collector_built()

C102 **Replacement mass per year (whole belt)**

Formula $\dot{m} = m_{belt} / L_{life}$

Values $2.11 \times 10^{21} / 50$

Result **4.22×10^{19} kg per year**

code: collector_design()

C103 **Fresh material over the mission**

Formula $m_{fresh} = m_{belt} + \dot{m}\ (1 - recycle)\ T$

Values $2.11 \times 10^{21} + 4.22 \times 10^{19} \times 0.05 \times 10{,}000$

Result **2.32×10^{22} kg = 7.0 % of Mercury's mass**

code: collector_design()

C104 **Manufacturing energy over the mission**

Formula $E = e_{emb}\ (m_{belt}(1 - b_0) + \dot{m}\ T)$

Values $e_{emb} = 1.00 \times 10^{8}$ J/kg

Result **4.23×10^{31} J = 0.0706 years of the belt's full electricity**

code: collector_design()

Tested mission: manufacturing energy charged 1.47×10^{31} J; panels built 1.47×10^{23} kg, fresh material 8.77×10^{21} kg.

6. Enabler E3 — space-mirror ring and simplified climate

Selected enabler: Space-mirror ring
A single mirror-ring technology reflects sunlight; the configured 1.6e15 m^2 film area can supply up to 1.10 x today's sunlight at 2.7 AU in the climate model. The agent sets the light level.
Why selected: Without added reflected sunlight, the simplified climate model falls into a cold state at the target orbit.
Gap to reality: Plain reflection is known physics, but the configured 1.6e15 m^2 and formation flight/pointing are far beyond industry; this is not a physical deployment design.
Modelled: Capacity (1/r^2), light sent, light turned away, mirror heat (energy ledger), mass and the simplified climate effect. The mirrors' real orbits, light pressure, pointing errors and illumination pattern are not simulated.

Selected enabler E3.1 - Space-mirror reflective field

Use: reflect sunlight onto Earth. Aluminium film, 5 g/m², reflectivity 90 %. Mirrors that are not needed are turned so their light passes Earth.

C105 **Mirror mass**
Formula $m = A\ \sigma_{areal}$
Values $1.60\times10^{15} \times 0.005$
Result **8.00×10^{12} kg**
code: mirror_design()

C106 **Area compared with Earth's disc; equivalent single disc**
Formula $A / (\pi\ R_{Earth}^2)$; $r = \sqrt{(A/\pi)}$
Values $1.60\times10^{15} / 1.28\times10^{14}$
Result **12.55 ×; one disc of radius 22,568 km, or 1,273,240 mirrors of 40 km**
code: mirror_design()

C107 **Area facing the Sun (45° tilt)**
Formula $A_{eff} = A \cos 45°$
Values $1.60\times10^{15} \times 0.7071$
Result **1.13×10^{15} m²**
code: mirror_design()

C108 **Largest light the mirrors can send to Earth, Earth at 1 AU**
Formula $P_{cap} = A_{eff}\ \rho\ S_{1AU} / r^2$
Values $1.13\times10^{15} \times 0.9 \times 1361.0 / 1^2$
Result **1.39×10^{18} W = 7.985 × the sunlight Earth intercepts (1.74×10^{17} W)**
code: mirror_design()

C109 **... Earth at the target (2.7 AU)**
Formula $P_{cap} = A_{eff}\ \rho\ S_{1AU} / a_t^2$
Values $1.13\times10^{15} \times 0.9 \times 1361.0 / 2.7^2$
Result **1.90×10^{17} W = 1.095 × today's sunlight on Earth**
code: mirror_design()

C110 **Mirror light needed at 2.7 AU to restore today's sunlight**

Formula $P_{need} = (1 - 1/a_t^2)\, S_{1AU}\, \pi\, R_{Earth}^2$

Values $(1 - 0.1372) \times 1.74\times10^{17}$

Result **1.50×10^{17} W = 0.788 of the mirror capacity (margin 26.9 %)**

code: climate_integrate()

C111 **Heat absorbed by the mirrors**

Formula $Q = (1 - \rho) \times$ sunlight on the mirrors

Values $1 - 0.9$

Result **10 % of the sunlight they catch (radiated to space)**

code: settle_interval()

C112 **Light pressure on a mirror at 2.7 AU (indicative, NOT simulated)**

Formula $a = 2\, S \cos^2 45° / (c\, \sigma_{areal})$

Values $2 \times 186.7 \times 0.5 / (2.998\times10^8 \times 0.005)$

Result **1.25×10^{-4} m/s² = 0.51 % of Earth's gravity at the mirror orbit**

Meaning Small but not negligible: the mirrors need active orbit and attitude control (separately classified support systems; not simulated).

code: indicative

Supporting system E3.2 - Terminator orbit and pointing

Use: the ring orbits Earth at 20 Earth radii in a plane at right angles to the sunlight, so each mirror always sees the Sun at 90° from Earth and is tilted 45°. The agent's action a4 sets how much light is aimed at Earth.

C113 **Orbit radius and period**

Formula $r = k\, R_{Earth}$; $P = 2\pi \sqrt{(r^3/(G\, M_{Earth}))}$

Values $r = 20 \times 6.371\times10^6 = 1.274\times10^8$ m

Result **P = 5.24 days**

code: mirror_design()

C114 **Size of the light spot on Earth from one mirror (Sun's angular size)**

Formula $D = \theta_{Sun}\, r$, $\theta_{Sun} = 2\, R_{Sun} / (a\ AU)$

Values θ_{Sun} at 2.7 AU = 0.00344 rad

Result **439 km at 2.7 AU (1,185 km at 1 AU)**

code: mirror_design()

C115 **Pointing accuracy to keep the spot on Earth (indicative)**

Formula $\delta\theta \approx R_{Earth} / r$

Values $6.371\times10^6 / 1.274\times10^8$

Result **0.0500 rad = 2.86° (a mirror turns by half that)**

code: indicative

Climate model (what the mirrors must achieve)

Two-box energy balance: a surface layer (atmosphere + ocean mixed layer) exchanging heat with the deep ocean; Earth's reflectivity rises from 0.30 to 0.60 as it freezes (snowball risk).

C116 **Effective emissivity (fixed greenhouse, 288 K today)**

Formula $\varepsilon_{eff} = S_{1AU}(1-\alpha_{warm})/4$ / (σ 288&sup4;)

Values 1361.0 × 0.70 / 4 / (5.670×10^{-8} × 288&sup4;)

Result **0.6105**

code: CLIM_EPS

C117 **Surface temperature equation**

Formula $C_s\ dT_s/dt = S(1-\alpha(T_s))/4 + q_w/(4\pi R^2) - \varepsilon\sigma T_s$&sup4; $- \gamma(T_s-T_d)$; $C_d\ dT_d/dt = \gamma(T_s-T_d)$

Values C_s = 8.0 W yr $m^{-2}K^{-1}$, C_d = 100.0, γ = 0.7 W $m^{-2}K^{-1}$, step ≤ 0.0625 y

Result **surface response time ≈ $C_s/(4\varepsilon\sigma T^3 + \gamma)$ = 2.0 years**

code: climate_integrate()

C118 **Equilibrium at 2.7 AU WITHOUT mirrors**

Formula solve $S(1-\alpha(T))/4 = \varepsilon\sigma T$&sup4;, $S = S_{1AU}/a_t^2$

Values S = 186.7 W/m² (0.1372 × today)

Result **152.4 K (-121 °C): snowball (frozen to the equator)**

code: climate_equilibrium()

C119 **Equilibrium at 2.7 AU with ALL mirror light on Earth**

Formula $S = S_{1AU}(1/a_t^2 + P_{cap}/(S_{1AU}\pi R^2))$

Values 0.1372 + 1.0954 = 1.2325 × today

Result **303.5 K — the agent aims less light (a set-point) to stay at 288 K**

code: climate_equilibrium()

C120 **Life indicators at the target temperature**

Formula $T(lat) = \bar{T} + \Delta T/3 - \Delta T \sin^2(lat)$, ΔT = 45 K; ice where T < 263 K; growth for 278-313 K

Values $\bar{T}$ = 288 K

Result **equator 303 K, pole 258 K, ice line 70.5°, growing area 74.5 %: temperate, like today**

code: life_indicators()

C121 **Tested mission: mirror light**

Formula sunlight on mirrors = to Earth + turned past Earth + heat

Values $8.20\times10^{28} = 1.04\times10^{28} + 6.34\times10^{28} + 8.20\times10^{27}$ J

Result **surface 288.4 K (largest deviation 2.6 K, 0 years outside the livable band); without mirrors it would be 157 K**

code: settle_interval(), summary()

7. Energy and mass ledger of the tested mission

Flow (J)	Historical pre-hold heuristic (high fidelity)
IN: sunlight on the collectors	1.25×10^{37}
IN: reserve used	0
IN: sunlight on the mirrors	8.20×10^{28}
reflected by the panels	1.25×10^{36}
collector heat	1.10×10^{37}
manufacturing (new + replaced panels)	1.47×10^{31}
panel electricity used	2.65×10^{35}
transmitter heat	5.30×10^{34}
microwaves sent	2.12×10^{35}
beam spill (missed the rectennas)	5.38×10^{33}
captured by the rectennas	2.07×10^{35}
rectenna heat	3.10×10^{34}
arm electricity	1.76×10^{35}
launcher draw	1.76×10^{35}
packet kinetic energy	8.78×10^{34}
launcher heat	8.78×10^{34}
mirror light onto Earth	1.04×10^{28}
mirror light turned past Earth	6.34×10^{28}
mirror heat	8.20×10^{27}
waste heat reaching Earth	1.24×10^{26}

C122 **Energy balance**

Formula: IN = reflected + collector heat + manufacturing + transmitter heat + spill + rectenna heat + packet KE + launcher heat + mirror flows

Values: 1.25385×10^{37} vs 1.25385×10^{37} J

Result: **relative residual 1.88×10^{-16} (round-off level: no energy created or lost)**

code: settle_interval() residual_rel

C123 **Efficiency of the beaming chain (panel electricity to arm electricity)**

Formula: $\eta = \eta_{mw}\,\bar{\eta}_{cap}\,\eta_{r}$

Values: 1.76×10^{35} / 2.65×10^{35}

Result: **66.3 %**

code: settle_interval()

C124 **Sunlight-to-packet efficiency**

Formula: $E_{\text{packet KE}}$ / $E_{\text{sun on used panels}}$

Values: 8.78×10^{34} / (2.65×10^{35} / 0.1654)

Result: **5.48 %**

Mechanical conservation (N-body, whole mission)	Relative residual
energy (system vs launches, escapes, impacts)	3.55×10^{-8}
momentum	1.90×10^{-11}
angular momentum	1.47×10^{-13}
mass	1.41×10^{-16}

Mass: 2.86×10^{22} kg ejected in 28,560 packets; 2.86×10^{22} kg left the Solar System; 1.00×10^{18} kg hit the Moon; 0 kg fell back to Earth; 0 kg hit Mars.

8. Historical pre-hold comparison and the current control task

Pre-hold archive only. The comparison below uses the old first-capture success definition. A reported PPO success or validated flag must not be interpreted as evidence that the current policy can maintain the target; these metrics are retained for provenance and have not been re-evaluated under the controlled-hold task.

HISTORICAL pre-hold high-fidelity mission (seed 2001)	No ejection	Hand-designed heuristic	Best PPO model
result	deadline	success	success
final a / e	1.0000 AU / 0.0122	2.6903 AU / 0.0072	2.6854 AU / 0.0145
simulated years (includes hold when recorded)	10,000	3,600	7,600
agent-controlled active hold	not recorded (pre-hold archive)	not recorded (pre-hold archive)	not recorded (pre-hold archive)
asteroid used	0.0 %	95.2 %	99.5 %
energy drawn	0 J	1.76×10^{35} J	1.73×10^{35} J
surface temperature	288.0 K	288.4 K	284.9 K
closest Mars pass	0.3670 AU	0.0502 AU	0.0470 AU
reward (scoring only, not physics)	-53.3	214.7	194.8

In the archived pre-hold test, the hand-designed strategy and PPO checkpoint reached the old target criterion. The PPO checkpoint is not evidence of current maintenance behavior; neither result is claimed to be optimal.

How the agent's five actions become physical commands

C125 **Action decoding (each action a_i in −1..1)**

Formula: $\text{throttle} = (a_0+1)/2$; $\theta = \pi\, a_1$; $\varphi = (\pi/2)\, a_2$; $u = u_{min} + (a_3+1)/2\,(u_{max}-u_{min})$; $S_{set} = 1 + 0.5\, a_4$

Values: u_{min} = 20,000 m/s, $u_{max} = 5.00\times10^6$ m/s

Result: **packet mass = throttle × $m_{pk,max}$; direction ($\sin\theta\cos\varphi$, $\cos\theta\cos\varphi$, $\sin\varphi$) in radial/along-track/normal axes; mirror set-point 0.5-1.5 × today's sunlight**

Meaning: The agent never sets Earth's velocity; it only commands packets and mirror aim, and physics does the rest.

code: OctoEarthEnv.decode_action()

C126 **Reward per decision (scoring only)**

Formula: $r = w_p\Delta\Phi - w_e(\text{e excess}) - w_i(\text{i excess}) - w_E E_{used}/E_0 - w_m m_{used}/m_0 - w_t - w_{Mars}\cdot\text{pen} - w_{clim}\cdot\text{pen}$ + active-hold progress/reset shaping + terminal terms

Values: w_p=100, w_E=10, w_m=5, w_t=0.02, w_{Mars}=20, w_{clim}=2, success 100, failure 100

Result: **a number for learning only; all physical results are reported separately**

code: OctoEarthEnv.step()

9. Selected enablers and supporting-system summary

Selected enabler / supporting system	Function	Configured performance (not a certification)
E1 selected · finite asteroid access	reaction mass	3.00×10^{22} kg (0.408 Moon masses)
E1 support · launch ring + 8 arms	throw packets	1.00×10^{18} kg at up to 5,000 km/s, 8/yr, η = 0.5; peak 6.34×10^{24} W
E1 support · power conditioning + reserve	launch energy	reserve 3.00×10^{35} J; beams first
E1 support · directional radiators	waste heat away from Earth	~1.84×10^{19} m² at 1,000 K; leak ≤ 1.00×10^{-9}
E2 selected · thin-film PV field	sunlight to electricity	5.00×10^{6} × 64,950 km, 2.11×10^{21} kg, η_{op} 16.5 % at 450 K
E2 support · microwave transmitters	electricity to beam	30 km arrays, 5.8 GHz, η 80 %, capture 86 %-100 %
E2 support · rectennas (8)	beam to electricity	1000 km, η 85 %; delivers 1.11×10^{25}-1.29×10^{25} W
E2 support · sail-tilt station keeping	hold slots	up to 9.08×10^{-5} m/s²; tested 730 km max drift
E2 support · clear sunlight lane	Earth sees the whole Sun	1.84×10^{-5} of the belt empty
E2 support · factories / feedstock	build + replace	4.22×10^{19} kg/yr; fresh 7.0 % of Mercury
E3 selected · space-mirror ring	reflect sunlight	1.60×10^{15} m², 8.00×10^{12} kg, ρ = 0.9; capacity 1.10 × at 2.7 AU
E3 support · terminator orbit + pointing	aim the light	20 R_{Earth}, 5.2 d period, pointing ~0.0500 rad
(not an enabler) Mars safety shield	keep Mars away	safe distance 0.0246 AU
(not an enabler) RL agent / control	decide every 100 y	5 actions, 31 observations

What is NOT modelled (honest list)

- **P5 Arm waste heat (~1e24 W, 6e6 x Earth's sunlight)** (open): The climate calculation assumes directional arm radiators with only a 1e-9 share reaching Earth. These are supporting equipment, not part of the selected E1 enabler. With an undirected radiator Earth would boil. *Remaining:* No physical radiator design, deployment or verified heat-rejection capability.
- **P6 Launch damage to Earth** (open): Each 1e18 kg packet leaves at ~2,500 km/s carrying ~3e30 J; shock, atmospheric loss and heating are not modelled. *Remaining:* Unresolved engineering hazard; non-damaging launch is not granted by an enabler or demonstrated by the model.
- **P7 Planetary context and long-term stability** (open): Mars, Mercury, Venus and Jupiter are modelled in both orbital engines, and their gravity is tested. The target orbit's long-term stability over geological times is not assessed; Saturn and planets beyond Jupiter are deliberately omitted. *Remaining:* Long-term stability and omitted outer-planet perturbations remain unverified.
- **P8 Day length, seasons, tides** (open): Spin, obliquity and tides are not modelled; the mirrors' illumination pattern on Earth is not modelled. *Remaining:* -

- **P10 Segment, hinge and launcher structural safety factors** (open): The participant guide requires every structural safety factor >= 1.5. Formula benchmarks are implemented, and p39 model 1 displays an assumed-capacity proxy, but no real material, cross-section, segment/hinge loads, buckling, self-weight, rotation or vibration analysis is connected to the main launcher. *Remaining:* No structural safety pass is claimed; material and geometry inputs are unspecified. The separate rulebook appendix thresholds were absent from the supplied PDFs.

10. What changes without a selected enabler or supporting system

These counterfactuals remove or reduce one selected enabler or one supporting system at a time. They show model consequences, not proof that any hardware can be built or that a single component is an enabler.

C127 **Without the asteroid reservoir**

Formula $\Delta v = v_\infty \ln(m_{sys} / (m_{sys} - m_{thrown}))$, $m_{thrown} = 0$

Result **Δv = 0: Earth cannot change its orbit (momentum is conserved; there is nothing to throw)**

code: feasibility()

C128 **Launcher limited to 30 km/s (fastest electromagnetic launchers studied)**

Formula $\Delta v_{cap} = \sqrt{(u^2 - v_{esc}^2)} \ln MR$

Values u = 30 km/s, MR = 1.004962

Result **138 m/s = 1.18 % of the required 11,658 m/s**

code: feasibility(max_ejection_speed_ms=3e4)

C129 **Without the energy reserve (energy_budget_j = 0)**

Formula $E_{avail} = P_{elec,min}\ T$ vs E_{min}

Values $1.11\times10^{25} \times 3.16\times10^{11}$ vs 1.66×10^{35}

Result **still enough: 3.52×10^{36} J available. At year 0 (30 % of the belt built) the arms get 3.87×10^{24} W vs 1.56×10^{24} W at full throttle with the strategy speed**

Meaning In the tested mission the reserve was not touched (all launch energy came from sunlight through E2). The reserve is an assumed store of energy with no source in the model; the mission does not need it.

code: feasibility(energy_budget_j=0)

C130 **Radiators that are not directional (~8 % of the heat reaches Earth)**

Formula $T = [(S(1-\alpha)/4 + q/(4\pi R^2)) / (\varepsilon\sigma)]^{1/4}$

Values $q = 0.08 \times 1.05\times10^{24}$ W

Result **T ≈ 8,296 K: Earth's surface would be destroyed**

code: climate balance (indicative)

C131 **Without the collector belt**

Formula E_{avail} = reserve only

Values 3.00×10^{35} J vs 1.66×10^{35} J

Result **passes only because of the assumed reserve; Earth's own sunlight for the whole mission is only 5.48×10^{28} J = 3.29×10^{-7} of the minimum**

Meaning Without E2 the launch energy has no physical source; the belt is what makes the energy come from sunlight.

code: feasibility(solar_collectors=False)

C132 Transmitters 10 times smaller (3 km)

Formula $\eta_{cap} = 1 - e^{-\tau^2}$, $\tau = D_t D_r/(\lambda d)$

Values D_t = 3 km (only the product $D_t D_r$ matters, so this equals a 10 times smaller rectenna)

Result **capture 16.8 % at 1 AU, 2.1 % at 2.7 AU (instead of 99.9 % / 86.3 %); the rest is spilled**

code: collector_power()

C133 Rectennas 10 times smaller (100 km)

Formula same Goubau relation

Values D_r = 100 km

Result **capture 16.8 % at 1 AU, 2.1 % at 2.7 AU: arm electricity 2.72×10^{23} W at the target**

code: collector_power()

C134 Without station keeping (tested, same mission)

Formula RK4 orbits of 12 collectors without the sail-tilt control

Values 3,600 simulated years

Result **largest drift 1.54×10^{8} km = 2,371 collector spacings (with it: 730 km): the belt would scramble, beams and lanes could not be kept**

code: BeltKeeper (check 19)

C135 Without the clear lane

Formula shade = 1 when $|z|/r < f$ (Earth's line of sight to the Sun crosses the band)

Values Earth in the ecliptic: $|z|/r \approx 0 < 0.3$

Result **shade = 1: Earth and its mirrors get no direct sunlight; only internal heat (~0.087 W/m²) would keep the surface near 40 K**

code: earth_shading()

C136 Without factories (belt stays at 30 %)

Formula $P_{elec}(b = b_0)$

Values b_0 = 0.3

Result **3.87×10^{24} W at 1 AU, 3.34×10^{24} W at 2.7 AU (still above the tested mean demand 1.82×10^{24} W, but aged panels would not be replaced: after 50 years the panels reach their end of life)**

code: collector_power()

C137 Without mirrors at the target

Formula equilibrium at $S = S_{1AU}/a_t^2$

Values 0.1372 × today

Result **152.4 K: snowball Earth (ice to the equator, no growing area)**

code: climate_equilibrium()

C138 Pointing error larger than Earth's angular size from the ring

Formula $\delta\theta > R_{Earth}/r$

Values 0.0500 rad

Result **the mirror light misses Earth: same as having no mirrors (C137)**

code: indicative

11. Technical data sheets: selected enablers and supporting systems

Each sheet identifies whether the item is a selected enabler or separately classified supporting hardware, its inputs and model performance, links to calculations, the counterfactual and its unresolved engineering gap.

E1.1 Finite asteroid mass reservoir (modeled as available feedstock) (Selected enabler E1)

Function	The selected assumption is access to a finite mass source; packets consume the configured reservoir.
Inputs	`asteroid_mass_kg = 3.00×10^22`
Configured performance	3.00×10^{22} kg (0.408 Moon masses); at least 1.41×10^{22} kg needed even at the maximum speed
Calculations	C20, C21, C22, C23, C127
Counterfactual	see C127
Unresolved engineering gap	Moving and docking a body of 40 % of the Moon's mass is far beyond any technology.

E1.2 Launch ring with 8 arms (mass driver) (Supporting system — not a selected enabler; associated with E1)

Function	Throws packets at the commanded speed and direction; all 8 arms throw one packet together.
Inputs	`max_packet_mass_kg = 1.00×10^18; max_ejection_speed_ms = 5.00×10^6; min_ejection_speed_ms = 20,000; ejections_per_year = 8.0; launch_efficiency = 0.5`
Configured performance	1.00×10^{18} kg packets, 2,480 km/s (strategy) up to 5,000 km/s, every 45.7 days; peak power 6.34×10^{24} W
Calculations	C24, C25, C26, C27, C28, C29, C30, C31, C32, C33, C34, C35, C36, C37, C38, C39, C40, C41, C42, C43, C44, C45, C46, C47, C128
Counterfactual	see C128
Unresolved engineering gap	No known structure; each launch carries ~10^{30}-10^{31} J.

E1.3 Arm power conditioning + energy reserve (Supporting system — not a selected enabler; associated with E1)

Function	Turns arm electricity into launch energy; beams first, reserve second.
Inputs	`energy_budget_j = 3.00×10^35`
Configured performance	handles up to 6.34×10^{24} W (7.92×10^{23} W per arm); reserve 3.00×10^{35} J (not used in the tested mission)
Calculations	C48, C49, C50, C129
Counterfactual	see C129
Unresolved engineering gap	Planetary-scale power electronics; the reserve has no physical source in the model.

E1.4 Directional arm radiators (Supporting system — not a selected enabler; associated with E1)

Function	Send launcher and rectenna waste heat to space, away from Earth.
Inputs	`arm_heat_to_earth_frac = 1.00×10^-9`
Configured performance	reject 1.05×10^{24} W; ~1.84×10^{19} m² at 1,000 K; leak to Earth ≤ 1.00×10^{-9}
Calculations	C51, C52, C53, C130
Counterfactual	see C130
Unresolved engineering gap	No radiator design; the leak fraction 1e-9 is assumed.

E2.1 Thin-film solar collectors (5 million) (Selected enabler E2)

Function	Intercept sunlight and make electricity.
Inputs	`collector_fraction = 0.3; collector_orbit_au = 0.5; collector_count = 5.00×10^6; collector_areal_density = 0.1; pv_efficiency = 0.25; collector_lifetime_years = 50.0`
Configured performance	2.11×10^{22} m², 2.11×10^{21} kg, 450 K panels, η_{op} 16.5 %, 1.90×10^{25} W of electricity at full output
Calculations	C63, C64, C65, C66, C67, C68, C69, C70, C71, C72, C73, C74, C75, C131
Counterfactual	see C131
Unresolved engineering gap	A Dyson-swarm-scale project (~10^{13} × all solar panels ever built).

E2.2 Microwave phased-array transmitters (Supporting system — not a selected enabler; associated with E2)

Function	Turn electricity into focused 5.8 GHz beams aimed at the arms; keep spill off Earth.
Inputs	`microwave_efficiency = 0.8; microwave_freq_ghz = 5.8; transmitter_diameter_km = 30.0; beam_spill_to_earth_frac = 1.00×10^-9`
Configured performance	30 km aperture, 3.04×10^{18} W each at full output, capture 86.3 %-99.9 %, pointing ~1.04×10^{-6} rad
Calculations	C76, C77, C78, C79, C80, C81, C82, C83, C84, C85, C132
Counterfactual	see C132
Unresolved engineering gap	Record power beaming is ~kW over ~km.

E2.3 Rectennas on the 8 arm tips (Supporting system — not a selected enabler; associated with E2)

Function	Receive the beams and turn them into electricity (only the arms receive power).
Inputs	`rectenna_diameter_km = 1000.0; rectenna_efficiency = 0.85`
Configured performance	1000 km each, 85 % efficient, up to 2.42×10^{12} W/m² (1.77×10^{9} × sunlight)
Calculations	C86, C87, C88, C89, C90, C91, C92, C93, C133
Counterfactual	see C133
Unresolved engineering gap	Beam intensity ~10^{8}-10^{9} × sunlight; no known receiver survives it.

E2.4 Sail-tilt station keeping (Supporting system — not a selected enabler; associated with E2)

Function	Hold each collector in its slot using part of its light push (no propellant).
Inputs	`collector_sk_authority = 0.5`
Configured performance	control up to 9.08×10^{-5} m/s²; tested effort 97 m/s per year, drift ≤ 730 km
Calculations	C94, C95, C96, C97, C98, C134
Counterfactual	see C134
Unresolved engineering gap	Solar sails demonstrated at m²-km² scale only.

E2.5 Clear sunlight lane (Supporting system — not a selected enabler; associated with E2)

Function	Keep a moving gap so Earth and Mars see the whole Sun.
Inputs	`earth_lane_clear = True`
Configured performance	1.84×10^{-5} of the belt kept empty for Earth, 3.27×10^{-5} for Mars
Calculations	C99, C100, C135
Counterfactual	see C135
Unresolved engineering gap	Coordinated motion of millions of collectors.

E2.6 Collector factories (Mercury material) (Supporting system — not a selected enabler; associated with E2)

Function	Build the belt up and replace aged panels.
Inputs	`collector_initial_built = 0.3; collector_build_years = 400.0; collector_recycle_frac = 0.95;` `collector_embodied_energy_jkg = 1.00×10^8`
Configured performance	4.22×10^{19} kg of panels per year; fresh material 7.0 % of Mercury
Calculations	C101, C102, C103, C104, C136
Counterfactual	see C136
Unresolved engineering gap	Self-replicating industry on Mercury is hypothetical.

E3.1 Mirror film panels (Selected enabler E3)

Function	Reflect sunlight onto Earth (or past it when not needed).
Inputs	`mirror_area_m2 = 1.60×10^15; mirror_reflectivity = 0.9;` `mirror_areal_density = 0.005`
Configured performance	1.60×10^{15} m², 8.00×10^{12} kg; must send up to 1.50×10^{17} W at 2.7 AU (capacity 1.90×10^{17} W)
Calculations	C105, C106, C107, C108, C109, C110, C111, C112, C137
Counterfactual	see C137
Unresolved engineering gap	Physically possible (plain reflection); demonstrated at 20 m (Znamya-2), needed ~10^{13} × that area.

E3.2 Terminator orbit + pointing (Supporting system — not a selected enabler; associated with E3)

Function	Keep the mirrors in sunlight at 20 Earth radii and aim the light (agent action a4).
Inputs	`mirror_orbit_re = 20.0; mirrors = True`
Configured performance	orbit period 5.24 days; pointing better than 0.0500 rad; light-pressure disturbance 1.25×10^{-4} m/s²
Calculations	C113, C114, C115, C138
Counterfactual	see C138
Unresolved engineering gap	Formation flying of about a million 40 km mirrors.

Mars safety shield (control software, not an enabler)

Use	Forecasts the Earth-Mars distance and overrides the agent's throw when Earth would come closer than the safe distance.
Calculations	C54, C55, C56, C57, C58, C59, C60, C61, C62
Gap to reality	Ordinary software; needs the Mars measurements of section 12.

RL agent PPO + heuristic baseline (control software, not an enabler)

Use	Chooses throttle, direction, throw speed and mirror light every 100 years from 31 observations.
Calculations	C125, C126
Gap to reality	Ordinary software; the tested numbers are in section 8.

12. Measuring instruments the control system needs (ordinary technology, not enablers)

The simulation gives the agent and the safety shield **perfect** knowledge of the state. A real system needs measurements. The instruments below are existing kinds of technology (no enabler needed); the required accuracies are *indicative*, derived from the mission tolerances (1-10 % of each tolerance). They are not simulated.

C139 **Earth's orbit size: required accuracy**

Formula $\delta a \le 1$ % of the tolerance; vis-viva: $\delta a = 2a^2 v\, \delta v / \mu \Rightarrow \delta v = \delta a\, \mu / (2\, a^2\, v)$

Values $\delta a = 0.01 \times 0.05$ AU = 74,799 km; a = 2.7 AU, v = 18,126 m/s

Result **position to ~74,799 km, velocity to ~1.68 m/s**

code: indicative

C140 **Earth-Mars distance: required accuracy**

Formula $\delta d \le 1$ % of the safe distance; forecast over the shield horizon

Values $0.01 \times 3.67 \times 10^6$ km; horizon 10 years

Result **relative position to ~36,746 km, relative velocity to ~0.116 m/s**

code: indicative

C141 **Global surface temperature: required accuracy**

Formula $\delta T \le 10$ % of the tolerance; $\delta S/S = \delta T / (dT/dS)$

Values 0.1 × 5 K; dT/dS ≈ 72 K per ×today (heuristic_climate)

Result **temperature to ~0.5 K; sunlight (incl. mirror light) on Earth to ~0.69 %**

code: indicative

C142 **Collector positions: required accuracy**

Formula $\delta x \le 1$ % of the tested drift with station keeping

Values 0.01 × 730 km

Result **~7 km per collector**

code: indicative

Quantity measured	Used by	Instrument (existing kind of technology)	Required accuracy
Earth's position and velocity (6 observations)	agent + shield	radio ranging/Doppler to spacecraft and stations (Deep Space Network type), planetary radar, star astrometry	C139
Orbit size, eccentricity, inclination	agent + success test	computed from the above (orbit determination)	e and i to ~0.0005 / 0.05°
Moon distance and whether it is bound	agent + failure test	lunar laser ranging (cm accuracy today)	~1 km is enough
Asteroid mass left	agent	counting and weighing packets (mass flow meters on the arms)	~1 % of a packet = 1.00×10^{16} kg
Energy reserve / beam power	agent + ledger	electrical power and energy meters on the rectennas and arms	~0.1 %
Mars position relative to Earth (5 observations)	agent + shield	radio tracking of Mars orbiters/landers, radar ranging	C140

Surface and deep-ocean temperature, sunlight on Earth (4 observations)	agent + climate test	satellite radiometers, ocean floats (Argo type), solar irradiance monitors	C141
Beam pointing at the rectennas	E2 transmitters	pilot beam from each rectenna (retrodirective phased array)	~1.04×10^{-6} rad
Collector positions in the belt	E2 station keeping	radio/optical ranging between neighbouring collectors + beacons	C142
Mirror attitude and aim	E3 pointing	star trackers + sun sensors on each mirror	better than 0.0500 rad

13. Appendix: every parameter

Key	Meaning	Value	Unit	Category
asteroid_mass_kg	Initial asteroid mass	3e+22	kg	Engineering assumption
max_packet_mass_kg	Max packet mass	1e+18	kg	Engineering assumption
max_ejection_speed_ms	Max ejection speed (exhaust velocity)	5000000.0	m/s	Engineering assumption
min_ejection_speed_ms	Min ejection speed	20000.0	m/s	Engineering assumption
energy_budget_j	Propulsion energy budget	3e+35	J	Engineering assumption
launch_efficiency	Launch efficiency	0.5	0-1	Engineering assumption
ejections_per_year	Ejection frequency	8.0	1/yr	Engineering assumption
target_au	Target orbit radius	2.7	AU	Mission requirement
mission_years	Mission duration	10000.0	yr	Mission requirement
decision_interval_years	Decision interval	100.0	yr	Simulation setting
tol_a_au	Tolerance on target radius	0.05	AU	Mission requirement
tol_e	Tolerance on eccentricity	0.05	-	Mission requirement
tol_inc_deg	Tolerance on inclination	5.0	deg	Mission requirement
validate_years	Target hold / passive stability duration	200.0	yr	Simulation setting
randomize_phase	Randomize starting position	True	on/off	Simulation setting
train_episodes	Training episodes	1000	-	RL hyperparameter
ppo_n_steps	PPO steps per update	1000	-	RL hyperparameter
ppo_batch_size	PPO mini-batch size	200	-	RL hyperparameter
ppo_epochs	PPO epochs per update	10	-	RL hyperparameter
ppo_learning_rate	Learning rate	0.0003	-	RL hyperparameter
ppo_gamma	Discount factor (gamma)	0.995	-	RL hyperparameter

eval_every_episodes	Evaluate every	50	episodes	RL hyperparameter
seed	Random seed	1	-	Simulation setting
w_progress	Reward: progress weight	100.0	points	Reward weight (scoring only)
w_eccentricity	Reward: eccentricity penalty	1.0	points	Reward weight (scoring only)
w_inclination	Reward: inclination penalty	1.0	points	Reward weight (scoring only)
w_energy	Reward: energy-use penalty	10.0	points	Reward weight (scoring only)
w_mass	Reward: mass-use penalty	5.0	points	Reward weight (scoring only)
w_time	Reward: time penalty	0.02	points/decision	Reward weight (scoring only)
w_success	Reward: success and hold	100.0	points	Reward weight (scoring only)
w_terminal_error	Reward: final-miss penalty	50.0	points	Reward weight (scoring only)
w_failure	Reward: failure penalty	100.0	points	Reward weight (scoring only)
w_mars_safety	Reward: Mars-proximity penalty	20.0	points	Reward weight (scoring only)
mars_gravity	Mars gravity on	True	true/false	Planetary safety (Mars)
other_planets	Mercury, Venus, Jupiter on	True	true/false	Planetary safety (Mars)
mars_max_tide_ratio	Max Mars tide on Earth (x Moon's tide)	0.01	-	Planetary safety (Mars)
mars_max_moon_perturbation	Max Mars pull on the Moon's orbit	0.001	-	Planetary safety (Mars)
mars_min_distance_km	Hard minimum Earth-Mars distance	2000000.0	km	Planetary safety (Mars)
mars_shield	Mars safety shield	True	true/false	Planetary safety (Mars)
shield_caution_factor	Shield caution factor	2.0	x safe distance	Planetary safety (Mars)
shield_horizon_years	Shield forecast horizon	10.0	yr	Planetary safety (Mars)
solar_collectors	Solar collectors on	True	true/false	Solar collectors (energy supply)
collector_fraction	Share of the Sun's output collected	0.3	0-1	Solar collectors (energy supply)

collector_orbit_au	Collector belt distance from the Sun	0.5	AU	Solar collectors (energy supply)
collector_count	Number of collectors	5000000.0	-	Solar collectors (energy supply)
collector_areal_density	Collector mass per area	0.1	kg/m^2	Solar collectors (energy supply)
pv_efficiency	Solar-cell efficiency	0.25	0-1	Solar collectors (energy supply)
microwave_efficiency	Electricity-to-microwave efficiency	0.8	0-1	Solar collectors (energy supply)
microwave_freq_ghz	Microwave frequency	5.8	GHz	Solar collectors (energy supply)
transmitter_diameter_km	Transmitter antenna diameter	30.0	km	Solar collectors (energy supply)
rectenna_diameter_km	Rectenna diameter (per arm)	1000.0	km	Solar collectors (energy supply)
rectenna_efficiency	Rectenna efficiency	0.85	0-1	Solar collectors (energy supply)
pv_temp_coeff	Solar-cell temperature coefficient	0.002	1/K	Solar collectors (energy supply)
collector_absorptance	Panel absorptance	0.9	0-1	Solar collectors (energy supply)
collector_emissivity	Panel emissivity	0.85	0-1	Solar collectors (energy supply)
pv_degradation_per_year	Panel degradation	0.002	1/yr	Solar collectors (energy supply)
collector_lifetime_years	Panel lifetime	50.0	yr	Solar collectors (energy supply)
collector_recycle_frac	Recycled share of replaced panels	0.95	0-1	Solar collectors (energy supply)
collector_initial_built	Belt built at year 0	0.3	0-1	Solar collectors (energy supply)
collector_build_years	Build-up time of the belt	400.0	yr	Solar collectors (energy supply)
collector_embodied_energy_jkg	Embodied energy of collectors	100000000.0	J/kg	Solar collectors (energy supply)
earth_lane_clear	Keep Earth's sunlight lane clear	True	true/false	Solar collectors (energy supply)
collector_sk_authority	Station-keeping authority	0.5	0-1	Solar collectors (energy supply)
climate_model	Climate model on	True	true/false	Climate + space mirrors

mirrors	Space mirrors on	True	true/false	Climate + space mirrors
climate_target_K	Target surface temperature	288.0	K	Climate + space mirrors
climate_tol_K	Temperature tolerance	5.0	K	Climate + space mirrors
mirror_area_m2	Total mirror area	1600000000000000.0	m^2	Climate + space mirrors
mirror_reflectivity	Mirror reflectivity	0.9	0-1	Climate + space mirrors
mirror_orbit_re	Mirror orbit radius	20.0	Earth radii	Climate + space mirrors
mirror_areal_density	Mirror film mass per area	0.005	kg/m^2	Climate + space mirrors
arm_heat_to_earth_frac	Arm waste heat reaching Earth	1e-09	0-1	Climate + space mirrors
beam_spill_to_earth_frac	Beam spill reaching Earth	1e-09	0-1	Climate + space mirrors
w_climate	Reward: climate penalty	2.0	points per 100 y	Reward weight (scoring only)
hifi_dt_hours	High-fidelity time step	6.0	h	Simulation setting
max_fragments	Max tracked fragments	48	-	Simulation setting
release_radius_m	Packet hand-over distance	2000000000.0	m	Simulation setting
fragment_escape_au	Fragment escape distance	60.0	AU	Simulation setting

142 calculations. End of report.